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A cloud-based adoption model for digital preservation services for university libraries

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Abstract

Purpose – This paper aims to review the critical determinants influencing the successful adoption of cloud computing as a solution to university libraries' challenges in providing efficient digital preservation services amid the emerging technologies in the 21st Century.

Design/methodology/approach – This study reviewed the literature concerning the in-depth revision of previous studies from the numerous published works to identify the factors on technology, organization and humans with the relevant constructs for our proper understanding of their influence on cloud computing adoption.

Findings – The literature review provides evidence that compatibility, availability and reliability; university libraries 'readiness; university management support; cloud migration strategy; staff perception; and attitude toward information technology change are significant drivers influencing cloud computing adoption.

Research limitations/implications – While this research provides valuable insights into cloud computing adoption within university libraries, it is essential to acknowledge certain limitations. First, this study is limited to university libraries, excluding other academic libraries of polytechnics and colleges and other types such as national, public and special libraries. Second, the focus is primarily on university libraries in the least developed and developing regions, particularly Africa and Asia. Third, the study relies heavily on literature review and theoretical analysis, potentially needing more empirical evidence or case studies to bolster its findings and recommendations.

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Practical implications – This study offers practical insights for university libraries seeking to enhance their digital preservation strategies through cloud-based platforms by identifying key barriers and facilitators. It also provides a cloud migration strategy as a roadmap or guide for university libraries to transition to the cloud easily. The practical model developed in this study and the cloud migration strategy bridges the gap between theory and practice, providing a guide for university libraries with the knowledge of cloud computing adoption for digital preservation services.

Originality/value – This study's primary value lies in its comprehensive examination of the challenges and potential solutions related to digital preservation in university libraries. It also developed a cloud-based model for university libraries, which will guide them in adopting cloud computing for efficient digital preservation services.

Keywords Cloud computing, Digital preservation, Technological factor, Organizational factor, Human factor, University libraries

Paper type General review

Introduction

Companies and organizations preserve their digital assets in a rapidly digitizing world. They preserve the content because it allows us to keep track of our history and culture more wholly and accurately than ever before so that the digital heritage is preserved as technology changes and file formats become obsolete (Sisinna, 2022). Just like organizations and other government agencies preserve cultural heritage and records, university libraries also preserve their scholarly outputs such as conference and seminar papers, historical documents, theses, dissertations, data sets, artifacts, journal articles (preprints and post-prints) and multimedia presentations (PowerPoint presentations, podcasts, images like photographs, video) to offer digital preservation services to users as part of the open-access (OA) initiative to address information access gaps (University of Southern Queensland, 2020). Therefore, this study is set to solve the data loss challenges facing university libraries, consequently hindering them from providing effective digital preservation services through OA initiatives.

Owing to the need for university libraries to fulfil their core mandate of providing access to digital content through digital preservation services, university libraries always strive to collect data generated from the members of university communities to preserve them for that purpose. The reason is that scholarly outputs in digital format, like other library resources, have assorted benefits compared to printed resources, as it is much easier, more user-friendly, efficient for searching and cross-referencing and space-saving and is freely accessible (Ashikuzzaman, 2024). As the need for digital content is increasing almost daily, university libraries are expected to preserve digital content effectively (Trianggoro and Prasetyadi, 2022). Amidst this exponential rate of data generation, however, preserving these digital resources becomes a challenge to university libraries, especially in the least developed and developing nations. Losing valuable material is expensive, and we need a sustainable safekeeping model for digital assets.

University libraries and other information centers adopt digital preservation strategies that include emulation, refreshing, migration, replication and development of locally hosted digital repositories (Gbaje, 2011; Ahmad *et al.*, 2023b). Despite the digital data preservation strategies mentioned above adopted by university libraries, there is a challenge still facing university libraries, especially in least developed and developing countries, as most universities in some developing countries in Africa and part of Asia operate on unreliable, obsolete and poor-performing information systems (IS) (Masenya and Ngulube, 2019; Ahmad *et al.*, 2023b; Ibrahim *et al.*, 2023).

These challenges facing these libraries include the inability to read the media because of technological obsolescence; the hardware and software sometimes are no longer supported

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or accessible; software-dependent problems and loss of the software programs that can interpret the digital object; a hardware device is no longer in production; the fragility of the media; virus attack; and natural disasters such as fire, flood, earthquake and tsunami. This storage system deteriorates rapidly, crashes quickly and is vulnerable to losing data, leading to the loss of treasures of scholarship. These frequent crashes translate into a digital loss of valuable university data and affect their information service delivery (Zubairu *et al.*, 2021).

Cloud computing as a proposed solution for digital preservation services

One emerging solution addressing digital data preservation challenges is cloud computing (CC) (Ibrahim *et al.*, 2023). CC delivers computing services – including servers, storage, databases, networking, software and analytics over the internet ("the cloud") to offer faster, innovative, flexible resources and economies of scale (Microsoft Azure, 2022). CC emerged in the digital environment to serve library users effectively and efficiently, especially internet-savvy users (Al-malah *et al.*, 2021) with cloud-based platforms, for example, such as Amazon S3 Glacier, Microsoft Azure Blob Storage, Google Cloud Archive Storage, CONTENTdm, Dropbox, DuraCloud, Preservica Cloud Edition Essentials (CEE) and Cloud Edition Professional (CEP), which are comprehensive cloud-based platforms. CEE or CEP stores its content within the Amazon Web Services storage system to hold digital assets (files) for fast, frequent access. University libraries will scale their storage without any hassle because the cloud has unlimited storage capacity; it solves the problem of hardware and software purchase, upgrade and maintenance.

Moreover, the cloud can preserve libraries' data properly, back up data in different locations, enhance visibility and provide 24/7 access to data uninterrupted (Ali *et al.*, 2022). CC also assists libraries in coping with budget cuts and constrained financial resources (Liu and Cai, 2023). Universities or libraries may have data recovery centers (DCR). However, it is costly to maintain such a center because it involves frequent software and hardware installation and upgrades, cooling systems and frequent training of a workforce to manage the center. It is better, safer and cost-effective for university libraries to store their data on the cloud to avoid equipment failure, site catastrophe or human malfeasance (Techtarget, 2024).

However, despite the accrued benefits of CC, higher education institutions (HEIs) and libraries, especially in least developed and developing countries, particularly in Africa and part of Asia, are moving slower than their counterparts in the developed world. CC adoption in these regions has been shallow, and the pace at which this technology adoption is languid (Qasem *et al.*, 2019; Bakri *et al.*, 2020) compared to enterprises (Supramaniam and Teoh, 2019; Eurostat Report, 2020; Amini and Javid, 2023). Even though few studies have been devoted to examining the factors that impact the adoption of CC in universities; but again, the examination has been at some segments or departments at the university level, for example, teaching and learning (Al-Ramahi *et al.*, 2022), university hospitals, university administration, university information technology (IT) department and university students personal information management (Alzaabi and Wahab, 2023). Only a little focus has been on the evolution of university libraries (Ibrahim *et al.*, 2023), specifically on the facility of CC adoption for digital data preservation services.

Furthermore, adopting CC in university libraries is influenced by multiple factors. Existing research has mainly centered on awareness and personal use of CC (Sudhir and Seena, 2018; Maina and Muthee, 2020). Scholars (Alharbil *et al.*, 2016; Lynn *et al.*, 2018; Qasem *et al.*, 2020; Gupta *et al.*, 2022) have mentioned some factors that are the cause of the low adoption rate of CC. However, this study focuses on three challenges affecting CC adoption. The first challenge concerns organizational factors; the organizational factor is the first issue. The biggest obstacle is getting top management's support because getting enough

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funding, internal resources, like IT-related resources, and finances to launch and maintain cloud services after launch are all critical issues to look into (Loukis *et al.*, 2017). Therefore, university libraries' adoption of CC must be evaluated in light of organizational factors.

The second challenge has to do with the technology aspect. IT managers must be familiar with the technology and its benefits and applications to business operations. As a result, the organization faces a significant hurdle in implementing cloud technologies. The human factor is the third challenge; it refers to an individual employee's innovativeness and qualities as a critical decision-maker, such as attitude toward IT change (Yusof *et al.*, 2008; Alharbil *et al.*, 2016). Existing literature has not provided sufficient insight into humans. This factor has to be explored to understand how the human component affects the adoption of new IT innovations in the library domain.

This study addresses this gap and takes an integrated approach by examining the connection of technological, organizational and human factors in adopting CC for digital preservation services in university libraries. The research novelty of this study lies in the holistic perspective, which offers a comprehensive understanding of the intricacies involved in technology adoption processes. Therefore, this study examines the effect of technology, organizations and human dimensions on CC adoption in university libraries for digital data preservation services.

Literature review

This study reviewed the literature on data curation, digital preservation and the impact of technological, organizational and human factors on adopting information technology. This study conducted an in-depth review of each theory or framework to evaluate its relevance in the context of CC adoption decisions. The technological adoption and acceptance theories used in this study include the Diffusion of Innovation (DOI) Theory, the Human Organization and Technology (HOT-fit) framework and the IS success model. Consequently, the chosen theories provide a solid foundation for examining the factors influencing the adoption of CC in university libraries.

Data curation in libraries

Data curation encompasses a range of actions undertaken to ensure that research data are fit for purpose and available for discovery and reuse and can help to improve the likelihood that data is more FAIR (Findable, Accessible, Interoperable and Reusable). The Data Curation Network has taken a collaborative approach to data curation, sharing curation expertise across a network of partner institutions and data repositories and enabling those member institutions to provide expert curation for a wide variety of data types and discipline-specific data sets (University of Pennsylvania Library, 2024). Data curation is vital because it involves managing, labeling and organizing data to ensure its quality, accessibility and usability (*Wells*, 2020). Tiwari (2018) contended that data is a new factor in production in this era. A massive volume of data production is leading to data deluge. The challenge is how to manage such data. Data curation is the data management activity that aims to enable the accessibility and reusability of the data. Data is managed in curation, preservation, archiving and storage layers.

Researchers are the data creators; these data can be reused if appropriately curated. Once the data is created, the job of data creators is over. Libraries are the experts in storing and managing the data in repositories (University of Pennsylvania Library, 2024). The data stored in the repository becomes junk if some value is not added (Tiwari, 2018). OCLC Research Library (2024) advises academic libraries to be prepared to respond to the need for data set management, supporting researchers' data needs during the research and providing

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ongoing preservation of and access to data sets. This activity will investigate topics related to data curation and libraries. The curation activity adds value to the raw research data, which can be further processed and reused. Libraries are the center of information management. For the management, organization and dissemination of information, libraries and information centers have used classification, cataloging and indexing techniques over the decades. Data curation deals with most of the activities that library professionals have been doing for ages. This expertise puts library professionals far ahead of other managers in data management, i.e. data curation. The understanding and expertise of the linked data, ontologies and metadata have brought a massive opportunity in data curation for library professionals (Freitas and Curry, 2016).

Tiwari (2018) submitted that the concept of data librarian is evident in stating the importance of library professionals in data management. Some American universities (e.g. the University of New Mexico) have data curation librarians who manage the whole data lifecycle. Universities like the University of Illinois, Toronto University, the University of North Carolina at Chapel Hill and many other institutions have already started courses on data curation for their master's curriculum of Library and Information Management. Data curation has many benefits.

Some of the benefits of data curation, as highlighted by Heidorn (2011), Freitas and Curry (2016) and Tiwari (2018), are as follows:

- data curation makes digital data reusable in the future;
- it intends to minimize the risk of data loss as the nature of digital data is fragile;
- data formats change regularly, and as a result, the available data can become corrupted. Data curation helps cope with this issue;
- it leads to easy acquisition and dissemination of the data; and
- it provides easy access and discovery of data to the users.

Digital preservation in libraries

Digital preservation is the series of managed activities (policies and actions) necessary to ensure continued access to digital materials or objects for as long as necessary (Wilson, 2017). In short, digital preservation means all the actions required to provide access to digital resources beyond media failure or technological change because it aims to take steps to ensure the longevity of electronic documents (Ahmad *et al.*, 2023b). It applies to documents created digitally and nondigitally born material that require digital preservation (those that come from the digitization process). Trianggoro and Prasetyadi (2022) described this as some people prefer to use the word 'archiving,' used in a computing science context to create a secure backup copy for a desired time. The digital preservation strategy is better achieved by moving information resources from old technologies to newer and better ones that resist any data corruption and hardware breakdown as they become available.

In libraries, collection managers may find digital preservation daunting, especially as collections grow in size, complexity and historical significance while new software, hardware, storage devices and data formats continually emerge. The rapid pace of technological advancement constantly threatens access to digital information, as inadequate preservation measures can lead to its loss. Risks associated with digital information encompass accessibility challenges and data loss (Ahmad *et al.*, 2023b; OCLC Research Library, 2024). Therefore, preservation has become critical to managing a library's most precious assets. Furthermore, the goal of any preservation program is to ensure long-term,

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ready access to an institution's information resources. Ruusalepp and Dobreva (2013) provided insights into digital preservation strategies, issues and challenges in data preservation, as shown in Table 1.

Table 1 summarizes the various strategies libraries and archive centers use to preserve digital data. However, these strategies often rely on specific combinations of hardware and software, which tend to become obsolete within a few years at best (Ruusalepp and Dobreva, 2013). The persistent evolution of digital technology poses a significant challenge to these preservation efforts. Information carriers are frequently unstable, deteriorating over a few years or decades, resulting in the inaccessibility of digital information. Challenges further arise from the difficulty in determining file formats, especially for older software and the potential risks of losing digital information resources in disasters such as fire, flood, equipment failure, viruses or direct attacks that turn off stored data and operating systems (Masenya, 2020). As previously mentioned, the overarching objective of any digital preservation system is to ensure continued accessibility of stored information over an extended period. Ideally, storage media should possess a prolonged life expectancy, exhibit high resistance to disasters, demonstrate sufficient durability to withstand frequent use and offer ample storage capacity. Consequently, there is a pressing need for an enhanced storage system like CC to facilitate efficient service delivery.

Diffusion of innovation theory

In 1962, Everett M. Rogers introduced the theory of the DOI through his work, Diffusion of Innovations, which is commonly referred to as Rogers' theory of technology adoption (Rogers, 2003). Affirmed the applicability of the DOI theory in studying the adoption of patient portals. Under the DOI theory, there are five established categories of technology adopters. These five categories of adopters fall under the elements of DOI. The five critical elements of this theory include Innovators, Early Adopters, Early Majority, Late Majority and Laggards (Rogers, 2003). While the Rogers's (2003) characteristics of innovation are relative advantage, complexity, compatibility, trialability and observability, As DOI only reflects the behavior of individuals in adopting new technological innovations, many researchers have combined DOI with other theories to describe the adoption process in organizations (Oliveira and Martins, 2011).

Technology–organizational–environmental framework

TOE framework was developed by Tornatzky and Fleischer (1990) to examine firm-level adoption of various IT products and services. It has become a general theoretical perspective on IT adoption (Gangwar *et al.*, 2014; Zhu *et al.*, 2006). When examining technology adoption, use and value creation from technology innovation, TOE is more accurate than other adoption models because it incorporates technological, organizational and environmental variables (Gangwar *et al.*, 2014). TOE is not restricted by industry or firm size. Therefore, it offers a comprehensive view of how users adopt technology, how it is implemented, how to anticipate problems, how it affects value chain activities, how it diffuses among organizations after adoption, what influences business innovation-adoption decisions and how to use technology to improve organizational capabilities (Salwani *et al.*, 2009; Lin and Lin, 2008; Zhu *et al.*, 2006).

The use of the TOE in this study makes sense because there has not been much research on organizational technology adoption using the framework, particularly in university settings (Alghushami *et al.*, 2020). The adoption of IT by organizations in developed and developing nations has been the subject of other research (Sallehudin *et al.*, 2018; Amini and Javid, 2023), with an emphasis on e-government, enterprise, e-readiness and technology

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Table 1. Digital preservation strategies: issues and challenges

Digital preservation strategies	Meaning	Issues and challenges
Refreshing	Making preservation copies from one storage medium to another of the same type, e.g. transferring data periodically from an old CD to a new one to avoid data loss due to media degradation	i. Quickly deteriorate or degrade ii. There is a risk that data loss will occur during the transfer of digital files from one generation of storage media to another
Emulation	Emulation is the strategy of imitating and replicating to overcome the technological obsolescence of hardware and software by emulating the digital object created. Emulation requires the creation of emulators, programs that translate code and instructions from one computing environment so they can be executed correctly in another	i. Obsolete digital objects may no longer be opened using current technologies
Replication or Bitstream copying	This strategy backs up data, making an exact duplicate of a digital object. It deals only with data loss due to hardware and media failure resulting from typical malfunction, decay, malicious destruction or natural disasters	i. Time: an internal team must set aside effort to implement and maintain a data replication system ii. Data loss is due to hardware and media failure resulting from typical malfunction, decay, malicious destruction or natural disaster
Migration	It involves periodically moving digital data from one hardware and software configuration to another and transferring content between formats and storage media. To prevent the physical storage medium and the data's encoding and format from obsolescence, it also entails converting data from one file format to another, such as MS Word to PDF or Open Document, from one operating system to another, such as Windows to Linux, or from one programming language to another, such as C to Java	i. A continuous task that needs to be repeated regularly ii. Migrating data to new formats as technology changes can be complex, time-consuming and expensive iii. As technology evolves, older file formats become obsolete and cannot be opened by newer software programs
Development of hosted digital repositories	Showcase and archive campus scholarly, creative works and other digital objects generated usually hosted on the on-premise server, also known as a local server, using software such as Dspace, Fedora common	i. The technical skills or expertise required to manage digital assets can be challenging to find and retain ii Frequent upgrade of the software and hardware technology

Source(s): Ruusalepp and Dobreva (2013)

adoption. It was, however, argued that Rogers's model should also be blended with other contexts or factors for a more holistic adoption approach. In line with this argument, this study found that the TOE framework includes the environment and organization contexts (not included in the DOI theory), thus becoming better able to explain intra-firm innovation adoption and, therefore, more complete. According to Oliveira and Martins (2011), the TOE framework has a solid theoretical foundation and application potential in IS/IT adoption.

Human organization technology (HOT-fit) framework

The HOT-fit framework, introduced by Yusof *et al.* (2008) as an assessment tool for health IS, offers a comprehensive approach to analyzing the relationship between humans, organizations and technology, as well as addressing issues, relationships and alignment within these components, and evaluating system performance and problems. It also effectively identifies information requirements in complex applications throughout the system development life cycle. Alharbil *et al.* (2016) extended the application of the HOT-fit framework to address human perspective issues encountered by IT staff in health-care organizations and high-performance computing (Lynn *et al.*, 2018).

Information systems success model

The IS success model (alternatively *IS success model* or *Delone and McLean IS success model*) is an IS theory that seeks to provide a comprehensive understanding of IS success by identifying, describing and explaining the relationships among six of the most critical dimensions of success along which IS are commonly evaluated. Initial development of the theory was undertaken by William H. DeLone and Ephraim R. McLean in 1992 and was further refined by the original authors a decade later in response to feedback received from other scholars working in the area (DeLone and McLean, 2003). IS success model has six dimensions, they are information quality, system quality, service quality, system use/usage intention, user satisfaction and net system benefits.

Research model development

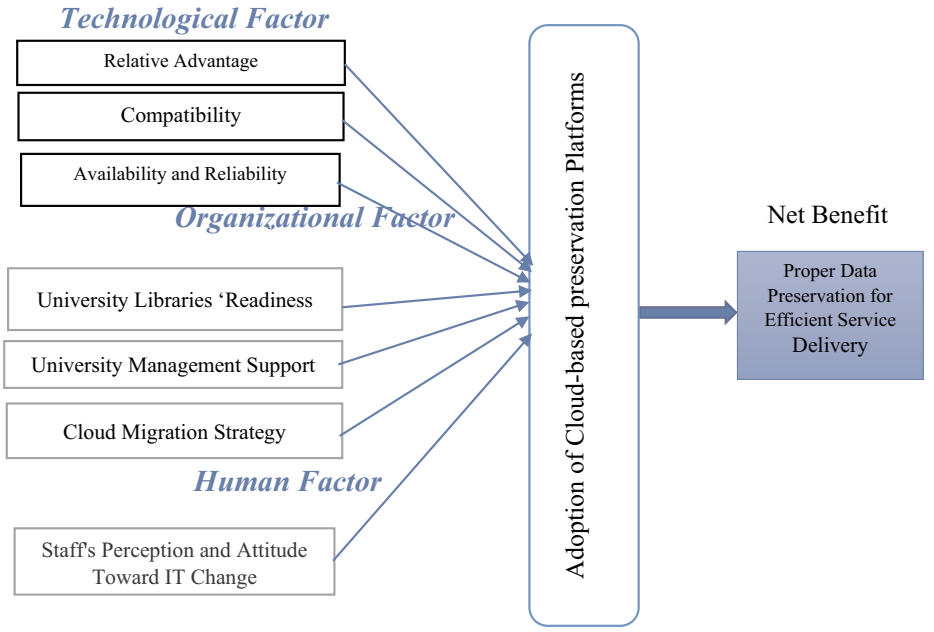
Based on the review, we proposed a cloud-based model for university libraries, as depicted in Figure 1. The study model is grounded on the relationship among some variables, with CC adoption as the dependent variable (DV). We integrate four theories to come up with the model. In the model, compatibility and relative advantage are DOI's construct and availability and reliability are the construct of Technology (T) in TOE. University Libraries 'readiness and university management support are the constructs under the Organization (O) factor in TOE; the cloud migration strategy is a new variable discovered from the literature related to organization migration strategy, so it is grouped under the organizational factor. Staff perception and attitude toward IT change is the construct under HOT-fit. The intentions to use CC as a system and the net benefits of the system are the constructs of the IS success model. The model developed will guide and encourage university libraries to speed up the adoption of CC for digital preservation to enhance the efficiency of service delivery. Figure 1 presents the research model.

Technological factors and the impact on cloud computing adoption

From this context, some elements considered for adopting CC are discussed below:

- *Compatibility of cloud computing*.

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Figure 1. Cloud-based adoption model for university libraries

Rogers *et al.* (2014) defined compatibility as the degree to which an innovation is perceived as consistent with the existing values, past experiences, and needs of potential adopters. The more CC platforms align with the internet platform of university libraries, the more university libraries develop an interest in using the benefits of CC and the more the possibility of reducing the degree of uncertainty among the university libraries. Also explained that cloud-based services should be compatible with the existing formats, interfaces and structured data. The study of Kandil *et al.* (2018) and Sathye *et al.* (2022) found that CC was compatible with the current systems used, and respondents indicated that integrating CC services was straightforward and compatible:

- *Relative advantage of cloud computing.*

Rogers *et al.* (2014) defined relative advantage as the extent to which a technology is perceived as providing more significant benefits for firms. It indicates that the current technology adopted is considered superior to the previous one due to anticipated advantages, such as reduced maintenance efforts, lower software costs and increased flexibility of computing resources libraries (Amini and Javid, 2023). In addition, the relative advantage of CC is associated with cost savings, potentially reducing overall operating costs by 25%–30% (Amini and Javid, 2023; Muzaffar *et al.*, 2023). There are less operating costs while using the cloud; university libraries deploying data to the cloud do not need to set up and manage their data centers. This significantly reduces overhead typically allocated for hardware and in-house IT. Furthermore, CC 1reduces upfront costs for new projects – CC models offer flexible payment options, like "pay-as-you-go," like payment for electricity bills, Airtime

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data subscription bills and other services that we pay as use, thereby reducing upfront investment in licensing and hardware (Imperva, 2023).

Availability and reliability of cloud computing. Major concerns in CC revolve around ensuring high availability and reliability. While the terms "reliability" and "availability" are often used interchangeably in the context of cloud services within a provider's data center, efforts have been made to distinguish between them. According to the Quality Glossary (2023), reliability is the likelihood that a product, system, or service, will perform its intended function adequately for a specified period without failure. It emphasizes the system's ability to fulfil its intended function as expected. On the other hand, availability, also known as dependability, refers to the likelihood that the system will be operational at any given time. For the quality of service, the selected cloud service provider is expected to guarantee 99.9% service level, uptime and availability, as emphasized by Okai *et al.* (2014) and Hassan *et al.* (2022). This requirement implies that the system must function correctly and be accessible whenever library users request. Interestingly, Kacha and Zitouni (2018) argued that data stored in the cloud is more secure than conventional networks because significant providers like Google, Amazon and Microsoft are better equipped to handle risks than on-premise servers or individual entities. Alghushami *et al.* (2020) conducted a study on factors influencing CC adoption in higher education institutions in Yemen, revealing a positive impact of reliability on CC adoption.

Organizational factors and the impact on cloud computing adoption

The variables under organizational factors and how they impact CC adoption are discussed below:

- *University Libraries' Readiness.*

The readiness of university libraries is measured by their level of awareness, resources, commitment and governance to embrace IT. This readiness encompasses the existing traditional IT systems and the financial investments in acquiring them (AlBar and Hoque, 2019; Ahn and Ahn, 2020). In this study, university libraries' readiness is determined by technological readiness within the library itself. Technological readiness involves both structural features and financial readiness. University libraries' financial readiness has been described with two dimensions, i.e. financial resources for CC implementation and ongoing expenses during usage. Finance and budget are other considerations for cloud decisions. Previous studies by Hwang *et al.* (2016), AlBar and Hoque (2019) and Ahn and Ahn (2020) affirmed that adopting CC necessitates adequate machine and IT infrastructure. This readiness encompasses the existing traditional IT systems and the financial investments made in acquiring them:

- *University Management Support.*

University management, or top management support, refers to an individual or a group responsible for making crucial organizational decisions. Defines top management support as the attitude of senior executives toward technology and their level of commitment to its implementation. Collectively, they decide whether to embrace new technologies and allocate resources to adopt the latest advancements (Van de *et al.*, 2016). Adopting information technologies in various management disciplines has emphasized the pivotal role of top management support in initiating, implementing and adopting technological innovations:

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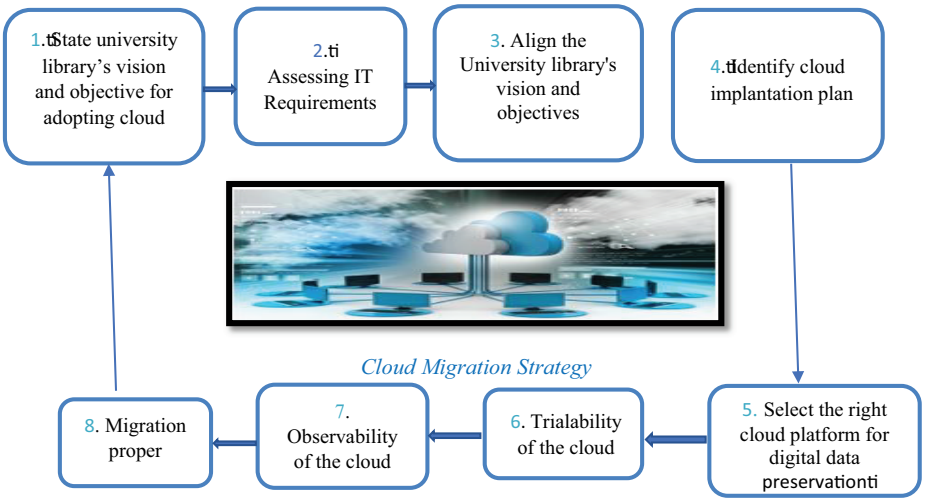
- *Cloud Migration Strategy.*

Cloud migration strategy is the plan that outlines the process of transferring or moving digital assets, such as data and applications, partially or wholly into the cloud environments (Avenga, 2021). Cloud migration can be done either by internal IT teams of trained professionals in systems administration, networking, programming and Web development or by hiring consultant companies as specialists. The most common type of cloud migration involves shifting organizational assets from on-premise data centers to the cloud (Avenga, 2021). However, there are other types; when organization assets are already located in the cloud, and the organization shifts the data to a different cloud, the process is called cloud-to-cloud migration. The most common type of cloud migration involves shifting organizational assets from on-premise to the cloud (Avenga, 2021). However, there are other types; when organization assets are already located in the cloud, and the organization moves the data to a different cloud, the process is called cloud-to-cloud migration.

This paper provides a cloud-based migration strategy for digital preservation services in university libraries. The strategy will help university libraries on how to migrate from the on-premise server to the cloud for cost saving and enhanced efficient service delivery. Figure 2 depicts the cloud migration strategy for university libraries.

The migration strategy is explained below:

- State University Library's vision and objectives for adopting the cloud: The cloud vision is a vision statement that gives the direction and purpose the university library wants to achieve. It motivates the staff and the cloud service provider who will work on cloud migrations and gives them a sense of direction (Imperva, 2023).
- Assessing IT requirements: This assessment aims to understand the IT requirements by defining their technical maturity and any factors needing improvement or reworking toward perfect cloud implementation. The IT assessment includes the



Source(s): Adapted from Atchison (2022) and Imperva (2023)

Figure 2. A proposed cloud migration strategy for university libraries

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current IT architecture, including power and an interface for users to access virtualized resources, among other requirements.

- Align the university library's vision and objectives: A cloud strategy supports organizational goals and objectives – what does the university library want to achieve through the cloud adoption journey? Is it in line with the university library's culture and overall vision? The vision and objective help the university library understand how it sees itself in the future and how different cloud environments can play a role in helping the university library reach its desired future state.
- Identify a strategy for cloud implementation: An explanation of the main actions and procedures the university library will take when putting a cloud solution into practice is called a cloud strategy roadmap. It offers a route map to assist the university library in achieving the intended results and aids in understanding the different tasks associated with the cloud implementation process. This implementation roadmap is part of the university library's cloud strategy, which also offers more details on how operations are now going: What is the current state of university operations? What challenges does the university library face, such as funding and IT requirements? How can the university library overcome these challenges? What library resources or data are currently used on-premise servers and must be moved to the cloud? When will your university library move the identified resources or data to the cloud?
- Select the right or suitable cloud platform for digital data preservation: This is an essential factor to be considered by the university library. The choice of the right cloud platform for digital preservation lies in the competency of the cloud service provider (CSP), which the university library should consider. These cloud platforms usually explore the use of CC technologies to test the perpetual access to digital content, allow institutions to provide data storage and access without having to maintain their dedicated technical infrastructure, explore strategies to help make content accessible in addition to delivering digital preservation services; they do not encrypt data but allow customers like university library choose to encrypt files before storing data in the cloud because they believe that it is the responsibility of the customer to maintain any encryption keys. According to Imperva (2023), the university library may quickly estimate cloud costs, including purchase and continuing operating costs, with a clear cloud strategy; the security features a library expects from a cloud solution; the scalability of cloud solution should offer; compliance and regulatory requirements cloud service provider should meet; and specific university library needs based on the needs of the university community the library serves.
- Trialability of the cloud: This is the ability of a university library to try out cloud-based platforms on a limited basis before making an adoption decision. This step enables university libraries to use CC for a while before deciding whether to adopt it (Al-Mascati and Al-Badi, 2016). Migrating to a cloud platform for university libraries needs investment, as most organizations and institutions might want to try the technology or innovation before making a final decision. Some cloud service providers give a free trial and a limited preview of the cloud preservation plus subscription plan. During the trial period, the customer account is free of charge for some hours or a few days, and at the end of the trial period, the customer's account will be canceled, and data will be automatically deleted. University libraries should start with a free trial account to experiment and later go for a subscription for between 6 and12 months as a trial from a reliable cloud service provider. University libraries may decide how to move their data to the cloud within the trial period, but it

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will be gradual. The reason is that things can get tense and complicated during a cloud migration. It may need to be clarified that everything will eventually smooth itself out, and migration will be a success. Especially if this is the first time migration has occurred and the cloud is new to the organization, like university libraries (Atchison, 2022). University libraries should also watch for the effects of innovation or migration within the next 6–12 months because the more easily an invention's outcomes can be seen, the more likely it is to be adopted and the more obvious its effects are to others.

- Observability of the cloud: University libraries' observation of the cloud migration will also help them think about technological innovation in a way compatible with existing values and needs to serve library users better and more efficiently. This trialability, observability and comparison for the compatibility period will enable the university libraries to decide whether to continue adopting CC.
- Migration proper: This is the final journey the university libraries embark upon, reaching a final destination after all the steps. It is hoped that the university libraries will move their data to a reliable cloud platform for proper preservation for efficient service delivery.

Human factor and the impact on cloud computing adoption

The human aspect encompasses decision-maker characteristics. The most common type of cloud migration involves shifting organizational assets from on-premise to the cloud (Avenga, 2021). However, there are other types; when organization assets are already located in the cloud, and the organization moves the data to a different cloud, the process is called cloud-to-cloud migration – adoption choices. Assessing the success of innovation adoption also relies on human actions (Yusof *et al.*, 2008; Jeyaraj *et al.*, 2006; Lynn *et al.*, 2018). The decisions of IT personnel play a pivotal role in shaping the adoption of IT innovation, as their knowledge and innovativeness can influence the process. According to Thong (1999), an individual's attitude toward technology is essential to adopting or rejecting technological innovation. In the context of university libraries, the human factor involves employee characteristics that contribute to the need and ability to embrace CC. The characteristics include IT prior experience and technology use support for using IT resources (Lian *et al.*, 2014; Haddad *et al.*, 2014; Alharbil *et al.*, 2016; Onayemi, 2022). The variables under the human factor and how they affect CC adoption are discussed below:

- *Staff perception and attitude toward IT change.*

Define attitude as a person's favorable or unfavorable expression toward an object or concept. Explore worker attitudes toward cloud technology, encompassing factors like attractiveness and the benefits of cloud technology on user performance and work ease. Schneider and Sunyaev (2016) contended that findings are scarce on the individual-level assessment of the decision to adopt new technology. It is crucial to understand the role of employees in university libraries in adopting CC. Khayer *et al.* (2020) emphasized the importance of considering how technology impacts individual employees, who act as decision-makers in determining technology adoption. According to Lindawati *et al.* (2023), individual characteristics, including employee behavior and responses to specific circumstances in the workplace, play a crucial role in the success or failure of technology adoption. Onayemi's study (2022) discovered that many respondents disagreed with adopting CC. This finding suggests that a considerable number of employees in Ugandan

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Small and Medium Entrepises (SMEs) approach CC with caution. It is crucial to understand the role of employees in university libraries in adopting CC.

Methodology

This study adopted the review methodology on CC and digital preservation services. The review method uses a systematic search strategy to find relevant literature across several databases, reducing bias and guaranteeing the inclusion of a wide range of viewpoints. Studies that have to do with CC adoption/implementation/utilization were searched irrespective of the sector or domain in relevant library databases and specialized search engines, including IEEE Xplore Digital Library, Sage Journals Online, ProQuest, Science Direct, Scopus, Emerald Insight, Springer Link, Web of Science and Google Scholars. The search terms included a variety of keyword combinations, such as "cloud computing adoption," "cloud computing implementation," "cloud computing utilization," "organizational factors," "technological factors", "human factors," and "perceived usefulness". Works such as conference papers, journal articles, book chapters and other published resources written in English were searched, retrieved, downloaded, reviewed and used in this study.

Findings

An extensive review of scholarly works shows that adopting CC is influenced by an assemblage of factors falling within three primary categories: technological, organizational and human factors. Furthermore, within these main categories, a range of variables, including compatibility, relative advantage, availability and reliability, university libraries' readiness, university management support, cloud migration strategy, staff perception and attitude toward IT, have been identified, each contributing to and shaping the dynamics of the adoption process.

Discussion of findings

The study aims to identify the determinants of CC adoption for digital preservation services in university libraries. The possible determinants were examined, namely, technological, organizational and human factors, as the independent variables, whereas cloud-based platforms were adopted as the DV.

Technological factors

Based on the research findings, technology significantly influences CC adoption, encapsulating three key variables: compatibility, relative advantages and availability and reliability. These factors collectively influence the decisions of university libraries regarding adopting CC.

The seamless integration of CC into library operations is facilitated when there is a high level of compatibility with existing systems. This assertion finds support in prior studies by Oliveira *et al.* (2014), Gangwar *et al.* (2015), Hsu and Lin (2016) and Al-Ramahi *et al.* (2022). This finding underscores the significance of assessing compatibility when introducing new library technologies. Recent literature, exemplified by Lindawati *et al.* (2023), studies the intention to adopt cloud-based accounting software in micro and medium enterprises. These studies have consistently highlighted the positive impact of compatibility on CC adoption across various organizations and enterprises.

Similarly, relative advantage has been found to correlate positively with the adoption of CC, leading to increased productivity or efficiency in service delivery, as evidenced by previous studies (Lian *et al.*, 2014; Gutiérrez *et al.*, 2015; Scholtz *et al.*, 2016; Hassan *et al.*, 2022; Amini and Javid, 2023). CC offers shared resources, enabling library users to access

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materials from any location, thereby saving time and money for organizations. Consequently, CC is poised to reduce costs and enhance the performance of university libraries.

Moreover, the adoption of CC hinges on the availability and reliability of the technology. Reliability pertains to the likelihood that CC will perform its intended function adequately for a specified duration without failure. Conversely, availability, also called dependability, denotes the likelihood that CC will be operational at any given time (Okai *et al.*, 2014; Hassan *et al.*, 2022). Availability plays a crucial role in an organization's adoption of CC as it offers online resources, allowing users to access the cloud anytime and from anywhere. Therefore, the system must operate smoothly and be accessible to library users around the clock (24/7) to ensure that information resources are available whenever needed. Previous studies (Safari *et al.*, 2015; Hsu and Lin, 2016) have indicated a positive relationship between availability, reliability and CC adoption.

Organization factor

In the realm of factors influencing the decision to adopt CC, organizational readiness stands out, encompassing the availability of financial and IT resources. This aspect has reaped attention in previous research, as indicated by its frequent mention in studies (Oliveira *et al.*, 2014; Gutiérrez *et al.*, 2015).

In the realm of factors influencing the decision to adopt CC, organizational readiness stands out, encompassing the availability of financial and IT resources. This aspect has reaped attention in previous research, as indicated by its frequent mention in studies (Oliveira *et al.*, 2014; Gutiérrez *et al.*, 2015), which underscores the impact of resource availability and reliability on CC adoption. In addition, university libraries must assess their current IT readiness and capabilities to determine whether they can successfully transition to CC and adopt cloud solutions. The finding shows the importance of having a suitable IT platform before adopting cloud technology.

Top management support, which this study terms university management support, constitutes another pair of influential constructs consistently associated with a positive impact on CC adoption. This association has been identified in studies by Lian *et al.* (2014), Gangwar *et al.* (2015), Hsu and Lin (2016). These factors hold particular significance for university libraries considering cloud adoption, as active prioritization and support from university management facilitate its implementation by allocating necessary funds and equipping university libraries with the required IT infrastructure.

Furthermore, cloud migration strategy plays a pivotal role in the adoption and success of CC. A study by Safari *et al.* (2015) suggests that an effective cloud migration strategy not only facilitates the transition to CC but also enables organizations to realize the full potential of cloud technologies in terms of cost savings, scalability, performance, innovation and agility while avoiding potential pitfalls in cloud migration. Previous studies such as have highlighted that the transition to CC services poses numerous challenges for Saudi Arabian SMEs. Among the immediate challenges is transitioning from existing technology to CC services. Saudi organizations, particularly those with extensive clientele, express apprehensions about migrating to cloud services due to potential disruptions and increased costs.

Human factor

Within the realm of human factors, the perceptions and attitudes of staff members toward IT innovations significantly influence the inclination to adopt CC. Previous studies (Alharbil *et al.*, 2016; Al-Ramahi *et al.*, 2022) underscore the pivotal role of staff mindset and readiness to embrace emerging technologies for the successful integration of CC in libraries. This observation resonates with findings from earlier research (Schneider and Sunyaev,

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2016), which emphasized that the success or failure of technology adoption hinges on employee responses to workplace dynamics. A comprehensive understanding and positive perception of CC among employees can spur thoughts of utilization, potentially enhancing productivity.

Implication

The study's significance is evident in both theoretical and practical areas.

Theoretical contribution

Theoretically, it enhances our understanding of technology adoption by evaluating the impact of various frameworks such as DOI, TOE, HOT-fit and IS success model constructs on CC adoption. Previous research on CC adoption, such as studies by Hassan *et al.* (2022) and Amini and Javid (2023), has predominantly focused on singular theories or frameworks like DOI or TOE, within specific domains. However, they still need to address certain technological factors sufficiently, particularly the reliability and availability of CC. Moreover, cloud migration strategies have received limited attention in CC adoption across various domains. Furthermore, human factors that have yet to be explored contribute significantly to theory-building, especially in the context of university libraries.

Practical implication

The study offers practical insights for university libraries seeking to enhance their digital preservation strategies through cloud-based platforms by identifying key barriers and facilitators. The study also provides a cloud migration strategy as a roadmap or guide for university libraries for easy transition to the cloud. Furthermore, this study developed a model based on the DOI–HOT-fit–IS success model. The practical model and cloud migration strategy bridge the gap between theory and practice, which can guide university librarians and other decision-makers with knowledge of CC adoption for digital preservation services.

Limitations and suggestions for future works

While this research provides valuable insights into CC adoption within university libraries, it is essential to acknowledge certain limitations. First, the study is limited to university libraries, excluding other academic libraries of polytechnics and colleges and other types such as national, public and special libraries. Second, the focus is primarily on university libraries in the least developed and developing regions, particularly Africa and Asia. Third, the study relies heavily on literature review and theoretical analysis, potentially needing more empirical evidence or case studies to bolster its findings and recommendations. Future research on CC adoption should encompass a broader spectrum of other types of libraries, including public, unique and national libraries.

Conclusions

This study investigates the determinants of CC adoption for digital preservation services in university libraries, shedding light on the factors and dynamics influencing CC adoption. The results highlight the need to adopt CC technologies for digital preservation services, providing a sustainable, scalable and affordable method of protecting priceless research outputs. The determinants such as compatibility, relative advantage, availability, reliability and staff perception and attitude toward IT change impact CC adoption. The study developed a model that adopted the DOI theory, HOT-fit, TOE and Information Systems Success (ISS) model. The adoption model described in this research provides university libraries looking to

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update their digital preservation plans with a strategy to migrate digital data to the cloud. By putting the suggested model of the CC and migration strategy into practice will ensure the lifespan and accessibility of research outputs, university libraries may contribute to the larger objective of preserving and expanding academic knowledge in the digital era.

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